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https://www.tensorflow.org/api_docs/python/tf/norm

> *Computes the norm of vectors, matrices, and tensors.*

Function Signatures

```
tf.norm( tensor, ord='euclidean', axis=None, keepdims=None, name=None )
```

Code Examples

```
tf.norm(
tensor, ord='euclidean', axis=None, keepdims=None, name=None
)
```


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tf.norm(
tensor, ord='euclidean', axis=None, keepdims=None, name=None
)
```


Documentation

Computes the norm of vectors, matrices, and tensors.

View aliases

Main aliases

tf.linalg.norm

Used in the notebooks

• Advanced automatic differentiation

- Matrix approximation with Core APIs
- Generate Artificial Faces with CelebA Progressive GAN Model
- Human Pose Classification with MoveNet and TensorFlow Lite
- Generalized Linear Models
- Neural machine translation with a Transformer and Keras

This function can compute several different vector norms (the 1-norm, the Euclidean or 2-norm, the inf-norm, and in general the p-norm for p > 0) and matrix norms (Frobenius, 1-norm, 2-norm and inf-norm).

Returns

Raises

numpy compatibility

Mostly equivalent to numpy.linalg.norm.
Not supported: ord <= 0, 2-norm for matrices, nuclear norm.
Other differences:
a) If axis is None, treats the flattened tensor as a vector
regardless of rank.
b) Explicitly supports 'euclidean' norm as the default, including for

higher order tensors.